Unspecific antecedents*

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1 Introduction

The similarity analysis of counterfactuals has trouble with disjunctive and otherwise unspecific antecedents. Alan Hájek has employed these troubles to advance his counterfactual skepticism and a chance-based approach to counterfactuals. I will argue that the data do not support these moves. The problem of unspecific antecedents, I will suggest, is better explained away as a pragmatic effect.

Let's briefly review the problem. According to the similarity analysis, a counterfactual $A > C$ is true iff C is true at all A worlds that are most similar, in certain respects, to the actual world.[1] As (Fine, 1975) and (Nute, 1975) first observed, the analysis fails to validate *Simplification of Disjunctive Antecedents* (SDA), the inference from $(A \vee B) > C$ to $A > C$ and $B > C$. To illustrate, consider an utterance of (1a) on a hot summer day.

(1) a. If it had rained or snowed, the match would have been cancelled.
b. $\rightsquigarrow$ If it had rained, the match would have been cancelled.
c. $\rightsquigarrow$ If it had snowed, the match would have been cancelled.

In this context, worlds where it snows are presumably more "remote" (less similar to the actual world) than worlds where it rains. All the closest worlds where it rains *or* snows are then worlds where it rains. By the similarity analysis, (1a) should imply nothing about what would have happened if it had snowed. Yet (1a) seems to imply (1c), as indicated by the curly arrows.

Much has been written about this objection. Some have argued that it calls for a different approach to the semantics of counterfactuals.[2] Others have suggested that the word 'or' in (1a) does not express Boolean disjunction.[3] Yet others have argued that the apparent entailment in cases like (1) is merely an implicature.[4] I will eventually side with a version of this last hypothesis. First, however, I want to explore a more direct line of response.

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[1] The analysis is commonly associated with (Stalnaker, 1968) and (Lewis, 1973). My formulation assumes the "Limit Assumption" but not the "Uniqueness Assumption". Neither is essential for the argument.

[2] See, for example, (Warmbrod, 1981) and (Fine, 2012).

[3] See, for example, (Lewis, 1977) and (Alonso-Ovalle, 2008).

[4] See, for example, (Loewer, 1976) and (Franke, 2011).

Friends of the similarity analysis have emphasized that the similarity standards that figure in the analysis need not match our intuitive judgements of overall similarity between worlds.[5] Above I assumed that (in the context of a hot summer day) snow worlds are more remote than rain worlds. If rain worlds and snow worlds are equally close, the similarity analysis predicts the inference in (1).

Any non-trivial version of the similarity analysis will still render SDA invalid. Unless all worlds are equally close, there will be instances of SDA in which one disjunct is more remote than the other, and these instances are predicted to be invalid. But this may not be a problem. After all, there appear to be counterexamples to SDA, such as (2), from (McKay and Van Inwagen, 1977).

(2) a. If Spain had fought with the Allies or the Axis [in World War II], it would have fought with the Axis.

b. $\not\rightsquigarrow$ If Spain had fought with the Allies, it would have fought with the Axis.

Perhaps the similarity analysis can be made to fit all the data, if only we plug in the right similarity measure? This hypothesis – although not expressed in these words – has recently been put forward by Alan Hájek.

Hájek also notes that SDA-type effects don't just arise with disjunctive antecedents:

(3) a. If it had rained, the match would have been cancelled.

b. $\rightsquigarrow$ If it had rained heavily, the match would have been cancelled.

(4) a. If Bob had wine, his medication would cause a severe reaction.

b. $\rightsquigarrow$ If Bob had red wine, his medication would cause a severe reaction.

In each of (1), (3), and (4), the 'a' conditional has an *unspecific* antecedent, leaving open, respectively, whether it rains or snows, how strongly it rains, and what kind of wine Bob consumes. The conditional with the unspecific antecedent appears to entail conditionals in which the antecedent is made more specific.

Hájek has invoked this generalized form of SDA to support his "counterfactual skepticism" – the claim that most ordinary counterfactuals are false. (See (Hájek, 2021b), (Hájek, 2021a), (Hájek, 2023)) Take (4a). Let's assume that Bob is on a medication that is guaranteed to cause a severe reaction if he consumes alcohol. In this context, we might accept (4a): his medication would cause a severe reaction if he had wine. But wait. What about de-alcoholised wine? That wouldn't trigger a reaction (let's assume). Since (4a) implies that Bob would have an allergic reaction no matter what type of wine he had, it is false. Or so Hájek would argue.

In general, ordinary counterfactuals often have unspecific antecedents. For them to be true, Hájek argues, they would have to remain true on every way

[5] See, for example, (Lewis, 1979, 42–48), (Stalnaker, 1984, 127ff.), and (Bennett, 2003, 195ff.).

of resolving the unspecificity. But we can often find unusual resolutions that would render them false.

I will return to this argument in section 7. First I want to explain how Hájek's chance-based version of the similarity analysis can account for inferences like (1), (3), and (4), as well as some other phenomena. I will then point at yet other facts – some well known, some new – that raise problems for Hájek's proposal, and indeed for any purely semantic treatment of SDA-type effects. I will sketch a (somewhat new) pragmatic derivation of these effects in section 6, before returning to the argument for counterfactual skepticism in section 7.

2 Chance-based similarity

I have stated the similarity analysis in abstract generality. We can be more concrete if we focus on conditionals whose antecedent describes a particular event, as in all the examples from the previous section.

It has often been observed that we seem to evaluate such counterfactuals by a "Rewind, Revise, Re-Run" heuristic: to evaluate whether $A > C$ is true, we consider worlds that are just like the actual world until shortly before the time of the antecedent ("rewind"), at which point the antecedent becomes true ("revise") and history unfolds again in accordance with the general laws of nature ("re-run"). The conditional is true iff C is true in all such worlds. (These are the "closest" A worlds to the actual world, in terms of the similarity analysis.)

How, exactly, does the antecedent become true, if it is not already true in the actual world? In a deterministic world, making a counterfactual antecedent true requires either a "miracle", as argued in (Lewis, 1979), or a divergence across the entire past, as argued in (Dorr, 2016). Let's avoid this difficult choice by assuming that the world is thoroughly indeterministic, so that a counterfactual antecedent describes a non-actual outcome of an actual chance setup.

Assume, then, that the antecedent A specifies an outcome of a chance setup S at a suitable *fork time* t.[6] Following the "Rewind, Revise, Re-Run" heuristic, we might say that $A > C$ is true iff the chancy laws of nature guarantee C on the condition that S yields A. More concisely: $A > C$ is true iff the chance of C given A at t is 1; that is, iff $Ch_t(C/A) = 1$, where Ch_t is the (actual) chance function at the fork time t.[7] Variants of this analysis have been defended by several authors throughout the years – including Skyrms (1980), Loewer (2007), Pearl (2000), Leitgeb (2012), and, most recently, Hájek (2023).

The chance analysis looks like a straightforward instance of the similarity analysis, albeit restricted to certain kinds of worlds and counterfactuals. To a first approximation, the similarity ordering would rank worlds that conform to the actual laws before worlds that don't, and sort worlds of the first kind by the

[6] See (Bennett, 2003, ch.13) for more on the notion of a fork.

[7] According to orthodox, Archimedean probability theory, events with chance 1 are not actually "guaranteed" to obtain. The present account seems to predict that if (say) a radium atom had been created today then it would not have decayed at x, for every future point in time x. Let's ignore this problem.

time until they first diverge from the actual world.[8]

One would probably want to finesse these criteria to balance lateness and smoothness of the divergence. Consider an example from (Bennett, 2003, 211): ‘If the German army had reached Moscow on 15 August 1941, …’. In the actual world, the Germans were far from Moscow at the dawn of 15 August. There was, however, a tiny chance that they would quantum tunnel to the Moscow outskirts. Since there were few Russian defenders in the city at the time, they would then almost certainly have encountered little resistance. But we don’t want to say that if the Germans had reached Moscow on 15 August then they would (probably) have encountered little resistance, or that they would have been baffled about how they got there. To make the right predictions, we need an earlier fork time that allows the Germans to reach Moscow in a less dramatic way.[9]

As presented above, the chance analysis also only covers counterfactuals in which the antecedent describes a single, specific event. What about counterfactuals of the form $(A \vee B) > C$,[10] where A and B describe different events? Some have proposed to extend the analysis by stipulating that $(A \vee B) > C$ is true iff $A > C$ and $B > C$ are both true.[11] This would validate SDA, and make the account incompatible with a similarity analysis. Hájek suggests that no special stipulation is needed: $(A \vee B) > C$ is true iff $Ch_t(C/A \vee B) = 1$, where t is the fork time for $A \vee B$.[12] What does this predict for the inference from $(A \vee B) > C$ to $A > C$ and $B > C$?

Suppose, to begin, that A and B are *fork-mates*, meaning that they have the same fork time t. (Perhaps they describe alternative outcomes of the same chance setup.) Plausibly, $A \vee B$ then has that same fork time. By the probability calculus, $Ch_t(C \mathbin{/} A \vee B) = 1$ entails $Ch_t(C \mathbin{/} A) = 1$ and $Ch_t(C \mathbin{/} B) = 1$, provided $Ch_t(A) \neq 0$ and $Ch_t(B) \neq 0$. That is, the SDA inference is truth-preserving whenever the disjuncts (and their disjunction) are fork-mates and both have a positive chance at the fork time.

In the examples from section 1, the disjuncts are plausibly fork-mates. Recall the match example:

(1) a. If it had rained or snowed, the match would have been cancelled.

[8] With this ordering, the similarity analysis implies that counterfactuals with true antecedents are true whenever they have a true consequent. If even such pseudo-counterfactuals are to be evaluated by conditional chance, we would need to replace the standard similarity semantics by an “antecedent-relative” semantics.

[9] Friends of chance-based or “interventionist” accounts sometimes suggest that their analysis is conceptually simpler and clearer than standard similarity semantics. (See, e.g., (Pearl, 2000, 239f.)) But the analysis is only simple and clear once we have identified the right causal model or the right chance function. Bennett’s example illustrates that this requires careful judgment.

[10] I use ‘$\vee$’ for classical disjunction. I generally assume that ‘or’ expresses $\vee$, setting aside the contrary view of, for example, (Lewis, 1977).

[11] Such a stipulation is often considered for the interventionist approach. See, for example, (Briggs, 2012), (Günther, 2017), (Ciardelli et al., 2018).

[12] This assumes that we can find a suitable chance function Ch_t that is defined for $A \vee B$. Pearl’s models do not generally provide such a function. I am not convinced that the chances of quantum physics could fill the gap. Let’s (again) ignore this problem.

b. $\rightsquigarrow$ If it had rained, the match would have been cancelled.

c. $\rightsquigarrow$ If it had snowed, the match would have been cancelled.

Physics tells us that the weather is thoroughly chancy. Even on a hot summer day, there is a small chance that it will start to snow. Both *rain* and *snow* therefore have positive chance, in the context of the example. On the chance account, this is enough to vindicate the inference in (1).

What about the case of (2), where we are not tempted to apply SDA?

(2) a. If Spain had fought with the Allies or the Axis, it would have fought with the Axis.

b. $\not\rightsquigarrow$ If Spain had fought with the Allies, it would have fought with the Axis.

A speaker who utters (2a) arguably believes that there was no chance that Spain would have fought with the Allies. Indeed, (2a) seems to convey just this belief. The chance account explains how: if there had been a chance that Spain fights with the Allies, (2a) would be false. And if $Ch_t(A)$ is 0, the chance account doesn't predict that $(A \vee B) > C$ entails $A > C$.

The chance account also seems to make good predictions about cases like (3) and (4), where an antecedent is unspecific but not disjunctive. Suppose A^+ is a more specific alternative to A. Whatever this means, it plausibly requires that A^+ entails A. By the probability calculus, $Ch_t(C \mathbin{/} A) = 1$ then entails $Ch_t(C \mathbin{/} A^+) = 1$, provided $Ch_t(A^+) \neq 0$. By the chance account, $A > C$ therefore entails $A^+ > C$ whenever A and A^+ are fork-mates and A^+ has a positive chance at the fork time. (SDA is a special case, where A is a disjunction and A^+ one of its disjuncts.)

With more than two resolutions of the relevant unspecificity, we can construct cases that combine elements of (1) and (2). Consider (5).

(5) If I had gone for a run today, I would feel less tired.

Suppose I usually go for a run either in the morning or in the early afternoon. Today I did not go for a run, and I feel tired. In this context, (5) seems to imply that I would feel less tired if I had gone for a run in the morning, and also if I had gone for a run in the early afternoon. It does not suggest that I would feel less tired if I had gone for a run at, say, *3 am*. The unspecific counterfactual does not seem to distribute over all resolutions of the unspecific antecedent, but only over the "realistic" ones.

The chance account might explain this effect. The "realistic" resolutions are those with positive chance at the fork time. Since the antecedent of (5) describes an interval that spans all of today, we may assume that the relevant fork time is at around the start of that day. When I utter (5), however, I arguably take for granted that there is no chance that I would have gone for a run at 3 am.[13]

[13] The case is similar to that of Pollock's coat, discussed in (Nute, 1980b, 104). Either case appears to challenge the similarity criteria proposed in (Lewis, 1979), which would favour later fork times even within the interval of time described by the antecedent. Under further scrutiny, these cases cast doubt on any version of the similarity analysis – as I'll explain at the end of the next section.

(5) draws attention to cases in which different resolutions of an unspecific antecedent have different fork times. Let's look at a disjunctive example:

(6) a. If I had gone for a run at 7am or 2pm, I would feel less tired.
b. $\rightsquigarrow$ If I had gone for a run at 7am, I would feel less tired.
c. $\rightsquigarrow$ If I had gone for a run at 2pm, I would feel less tired.

The inference looks good, even though the fork time for (6b) is arguably different from that for (6c). We can still vindicate the inference if we assume, as above, that the fork time t for (6a) lies at or before the interval covered by the antecedent, so that it coincides with the fork time for (6b). Let t' be the later fork time for (6c). Assuming that $Ch_t(C \ / \ A)$ and $Ch_{t'}(C \ / \ B)$ are not zero, $Ch_t(C \ / \ A \vee B) = 1$ entails $Ch_t(C \ / \ A) = 1$ and $Ch_t(C \ / \ B) = 1$. Under plausible assumptions about the dynamics of chance[14], $Ch_t(C \ / \ B) = 1$, in turn, implies $Ch_{t'}(C \ / \ B) = 1$.

I have one more piece of good news, before I bring in the bad news.

It is often said that alleged counterexamples to SDA like (2a) feel unnatural or marked. As (Lassiter, 2018) points out, they become more natural if we insert a probability operator:

(7) a. If Spain had fought with the Allies or the Axis, it would probably have fought with the Axis.
b. $\not\rightsquigarrow$ If Spain had fought with the Allies, it would probably have fought with the Axis.

This is a challenge for accounts that treat SDA as unrestrictedly valid. It is not a problem for the chance account.

To apply the chance account, we first need to know how it extends to 'would probably' counterfactuals. A natural idea is that 'if A, would probably C' is true iff the relevant chance of C conditional on A exceeds some contextual threshold.[15] That (7a) does not entail (7b) then follows from the fact that $Ch_t(C \ / \ A \vee B) > \theta$ does not entail $Ch_t(C \ / \ A) > \theta$, even if both disjuncts have a positive chance at t.

In sum, Hájek's chance account seems to reconcile the similarity analysis with our judgments about counterfactuals with unspecific antecedents.

Alas, things are not that simple. Let's have a look at some more data.

[14] We need to assume that the chance of an event conditional on another cannot decrease from 1. Informally, if the state of the world at t entails that the only possible outcome of *run at 2 pm* at t' is C then this state of the world can't evolve into one where there is a possible outcome of *run at 2 pm* other than C at t'.

[15] One way to derive this interpretation, drawing on the classical theory of (Kratzer, 2012), assumes that the 'if' clause restricts a modal: in ordinary counterfactuals, that modal (possibly lexicalized by 'would') expresses chance 1, in 'would probably' counterfactuals, it expresses high chance. Assuming that probability modals get "restricted" by conditionalization (compare (Yalcin, 2010)), it follows that 'if A, would C' expresses $Ch_t(C/A) = 1$, while 'if A, would probably C' expresses that $Ch_t(C/A)$ is high.

3 Context and content

Return once more to the match case (1). According to the chance account, the inference from (1a) to (1c) is licensed by the fact that even on a hot summer day there is a positive chance of snow. But does the inference really rest on this non-obvious fact about chance? I suspect that hearers would be inclined to infer (1c) from (1a) even if they don't think the weather is so thoroughly chancy. Counterfactuals like (1a) seem to support SDA even if one of the disjuncts is assumed to have chance zero.

For a more controlled example, imagine I'm about to draw a ball from an urn. You bet that I'll draw a red ball. The ball I draw is green. In this context, (8a) seems false, as it appears to imply (8b).

(8) a. If I had drawn a red ball or a yellow ball, you would have won.
b. $\rightsquigarrow$ If I had drawn a yellow ball, you would have won.

The apparent entailment remains intact, I think, even after we have both inspected the urn's content and seen that it does not contain any yellow balls, so that (as I hereby stipulate) there was no chance of drawing a yellow ball.

In this kind of case, one can, however, also utter a "Spain" type conditional. Having established that there were no yellow balls in the urn, I might felicitously assert (9).

(9) If I had drawn a red ball or a yellow ball, I would have drawn a red ball.

Curiously, after (9) has been accepted, (8a) becomes more acceptable: it no longer seems to imply (8b).

This curious effect has been noted in (Nute, 1980a). Nute observes that we would normally reject (10), as it suggests that Hitler would have been pleased no matter which side Spain had joined.

(10) If Spain had fought with the Allies or the Axis, Hitler would have been pleased.

(10) becomes more acceptable, however, if it follows (2a).

Needless to say, the chance account does not explain this context-sensitivity. If we wanted to account for the present data in terms of the similarity order, we would have to assume that possibilities can become "close" simply by being mentioned, even if they have zero chance, and that this effect can be blocked by utterances like (9) or (2a).[16]

Here is a somewhat different, but related problem. Compare (10) with (11).

(11) If Spain had joined the war, Hitler would have been pleased.

Modulo some facts that are common ground, the two antecedents are equivalent. Yet (11) is more acceptable than (10). Unlike (10), it does not suggest that Hitler would have been pleased no matter which side Spain had joined. What explains the difference?

[16] (Nute, 1980a) proposes something like this.

Hájek might argue that (10) and (11) are both false, and that (11) looks more acceptable because the falsity is easier to miss. But take a different example.

(12) If Trump had won in 2020, he would have won a second consecutive term.

This is plausibly true, even by Hájek's skeptical standards. Trump's winning in 2020 would have *constituted* winning a second term. No unlikely chance events could have intervened between the antecedent and the consequent, for they describe the same event. But they do so only on the background of what happened in 2016. Thus (13) is false:[17]

(13) If Trump had lost in 2016 and won in 2020, he would have won a second consecutive term.

Now consider (14).

(14) If Trump had won in 2020 after either winning or losing in 2016, he would have won a second consecutive term.

This also seems false, as it seems to imply (13). But its antecedent is (practically) equivalent to that of (12).[18]

Another problem. In the previous section, I argued that the chance account can predict the SDA effect in (6), where the disjuncts have different fork times.

(6) a. If I had gone for a run at 7am or 2pm, I would feel less tired.
b. $\rightsquigarrow$ If I had gone for a run at 7am, I would feel less tired.
c. $\rightsquigarrow$ If I had gone for a run at 2pm, I would feel less tired.

The explanation assumed that the fork time for (6a) coincides with that of (6b) and not with the later fork time of (6c). This is a natural assumption. But it has an unwelcome consequence. We now can't predict the converse inference from (6b) and (6c) to (6a). No independently plausible assumptions about chance would ensure that $Ch_t(C/A) = 1$ and $Ch_{t'}(C/B) = 1$, with $t < t'$, together entail $Ch_t(C/A \vee B) = 1$. Yet the inference from (6b) and (6c) to (6a) looks at least as safe as the SDA inference from (6a) to (6b) and (6c).

This problem does not just arise for the chance account. Any attempt to generalize the "Rewind, Revise, Re-Run" approach to temporally unspecific antecedents like (5) and (6) faces a dilemma. To predict the SDA-type effect, one must assume that the fork time for a temporally unspecific antecedent can

[17] We have here a counterexample to Antecedent Strengthening. Do we also have a counterexample to the generalized form of SDA? I wouldn't say so. The hypothesis that Trump won in 2020 – the antecedent of (12) – is not *unspecific* with respect to the 2016 election. It isn't about the 2016 election at all.

[18] One might try to account for the difference between (12) and (14) in terms of the relevant fork time: perhaps the fork time for (14) is before 2016, simply because that year is mentioned in the antecedent, while the fork time for (12) is later. I doubt that this gets to the heart of the matter. See (Champollion et al., 2016) and (Romoli et al., 2022) for other cases in which the formulation of the antecedent seems to make a difference.

be as early as the fork time for its earliest resolution. If later resolutions can have later fork times, however, this invalidates the converse of SDA.[19]

At this point, the prospects for reconciling SDA-type effects with the similarity analysis no longer look as rosy as they did in the previous section. But the really bad news is still to come.

4 Beyond 'would' counterfactuals

So far, we have looked at 'would' counterfactuals with disjunctive or otherwise unspecific antecedents. SDA-type effects also arise for 'might' counterfactuals (as (Alonso-Ovalle, 2006) notes):

(15) a. If it had rained or snowed, the match might have been cancelled.
b. $\rightsquigarrow$ If it had rained, the match might have been cancelled.
c. $\rightsquigarrow$ If it had snowed, the match might have been cancelled.

Let's see if the chance account can explain this. On the chance account, it's natural to assume that a 'might' counterfactual $A >_m C$ is true iff $Ch_t(C/A) > 0$, where t is the relevant fork time. This makes 'might' and 'would' conditionals duals of one another.[20] But it does not predict the inference in (15): $Ch_t(C \ / \ A \vee B)$ can easily be positive even though $Ch_t(C \ / \ B)$ is zero.

There's no easy way out. Suppose we accept SDA for both 'would' and 'might' counterfactuals, perhaps conditional on certain facts about chance or whatever. Assuming duality, we can then derive $B > C$ from $A > C$:[21]

1.	$A > C$	(Assumption)
2.	$\neg(A >_m \neg C)$	(1, Duality)
3.	$\neg((A \vee B) >_m \neg C)$	(2, SDA for 'might' counterfactuals)
4.	$(A \vee B) > C$	(3, Duality)
5.	$B > C$	(4, SDA for 'would' counterfactuals)

We can similarly derive $B >_m C$ from $A >_m C$:

1.	$A >_m C$	(Assumption)
2.	$\neg(A > \neg C)$	(1, Duality)
3.	$\neg((A \vee B) > \neg C)$	(2, SDA for 'would' counterfactuals)
4.	$(A \vee B) >_m C$	(3, Duality)
5.	$B >_m C$	(4, SDA for 'might' counterfactuals)

We could avoid these triviality results by dropping the duality assumption. According to a popular alternative, 'might' counterfactuals should be understood as 'would' counterfactuals embedded under 'might'.[22] This is not an

[19] The converse of SDA is valid in standard similarity semantics such as (Lewis, 1973) and (Stalnaker, 1968). The natural assumption that temporally unspecific antecedents can have fork times that precede the fork time of their latest resolutions is therefore incompatible with standard similarity semantics. It leads to an "antecedent relative" similarity semantics. (See, e.g., (Stalnaker, 1984, 129ff).)

[20] Strictly speaking, we only predict that $\neg(A > C)$ and $A >_m \neg C$ are equivalent *if* $Ch_t(A) \neq 0$.

[21] See (Cariani and Goldstein, 2020, 11f.) for a similar result.

[22] See, for example, (Stalnaker, 1980), (DeRose, 1999), (Williams, 2010).

attractive option for the chance account: 'if the coin were tossed, it might land heads', for example, can surely be true even if the coin is certain to be fair, so that the conditional chance of heads is definitely not 1. To yield sensible verdicts about such cases, the analysis of $A >_m C$ as $\Diamond(A > C)$ is usually combined with a view of 'would' counterfactuals that validates "Conditional Excluded Middle":

(CEM) $\quad (A > C) \vee (A > \neg C)$.

This alternative approach, however, is also incompatible with the validity of SDA, as the following deduction shows. 'CNC' in the second step stands for 'Conditional Non-Contradiction': $A > C$ and $A > \neg C$ cannot both be true. This is widely accepted, at least if A is itself possible.[23]

1.	$A > C$	(Assumption)
2.	$\neg(A > \neg C)$	(1, CNC)
3.	$\neg((A \vee B) > \neg C)$	(2, SDA)
4.	$(A \vee B) > C$	(3, CEM)
5.	$B > C$	(4, SDA)

It gets worse. Yet other types of conditionals also give rise to SDA-like effects. Of special interest are deontic conditionals, as in (16).

(16) a. If you consume alcohol or cocaine, you ought not to drive.
b. $\rightsquigarrow$ If you consume alcohol, you ought not to drive.
c. $\rightsquigarrow$ If you consume cocaine, you ought not to drive.

According to an attractive and popular analysis of deontic conditionals, 'if A, ought C' is true (roughly, and on its most salient reading) iff the best of the circumstantially accessible A worlds are C worlds, where "circumstantial accessibility" is determined by relevant facts and "best" is evaluated in accordance with relevant rules or norms.[24] This is structurally analogous to the similarity analysis of 'would' counterfactuals, and it does not validate SDA.

Concretely, suppose consuming alcohol is less bad than consuming cocaine by the relevant rules or norms, and both possibilities are circumstantially possible. Then the best (accessible) worlds where you consume alcohol or cocaine are worlds where you consume alcohol. In this context, (16a) should imply nothing about what you should do if you consume cocaine. But it does: it seems to imply (16c).[25]

An SDA-type effect can also be observed with plural descriptions – which have been argued to be structurally analogous to counterfactuals.[26]

[23] Again, see (Cariani and Goldstein, 2020) for a series of similar results.

[24] See, for example, (Feldman, 1986, ch.4) and (von Fintel and Heim, 2021).

[25] As in the case of 'would' counterfactuals, this SDA effect seems to generalise beyond disjunction; it arises with 'may' as well as 'ought'; and it can be blocked by moves analogous to that in (2). For example, suppose I am determined to consume either alcohol or cocaine before going on a drive. You'd rather that I consume neither. If you think that cocaine is even worse than alcohol, you might still say: 'if you're going to consume alcohol or cocaine, you ought to consume alcohol'. This does not suggest that if I consume cocaine then I should (in addition) consume alcohol. (Compare the "gentle murder paradox" of (Forrester, 1984).)

[26] See, for example, (Schlenker, 2004)).

(17) a. The poorest countries once colonized by France or Spain are all in Africa.
b. $\rightsquigarrow$ The poorest countries once colonized by France are all in Africa.
c. $\rightsquigarrow$ The poorest countries once colonized by Spain are all in Africa.

The inference should be invalid: if we list the countries once colonized by France or Spain, we might find that the poorest ones have all been colonized by France. (In fact, this seems to be the case.) The fact that these countries are all in Africa should then tell us nothing about the location of the poorest countries once colonized by Spain. Why, then, does the inference in (17) look OK?

Finally, it has often been noted[27] that the SDA effect closely resembles the "Free Choice" effect that arises when disjunctions are embedded under possibility modals, as in (18).

(18) a. You may have tea or coffee.
b. $\rightsquigarrow$ You may have tea.
c. $\rightsquigarrow$ You may have coffee.

According to the standard semantics of possibility modals, (18a) is true as long as at least one of tea and coffee is allowed. It should not entail (18b) or (18c).

There appears to be a general mechanism by which constructions of the form $\phi(A \vee B)$ can be used to convey $\phi(A)$ and $\phi(B)$, even though this is not supported by the literal meaning of these constructions. The chance account may explain why a 'would' counterfactual $(A \vee B) > C$ can be used to convey $A > C$ and $B > C$, but the explanation does not carry over to 'might' counterfactuals, to deontic conditionals, or to any of the other constructions I've mentioned.

Of course, it is possible that the SDA effect for 'would' counterfactuals can be explained in terms of the semantics of 'would', while something else – some pragmatic story, perhaps – explains the apparently similar effect for other constructions.[28] There are, however, reasons to think that the same mechanism is at work in all these cases. More concretely, there are reasons to think that the mechanism in all the cases I've reviewed, and especially in the case of 'would', involves a scalar implicature.

5 The implicature hypothesis

Utterances of a comparatively weak sentence often convey that a stronger alternative is false, as in (19).

(19) a. It's possible that it will rain.
b. $\rightsquigarrow$ It's not certain that it will rain.

[27] for example, in (van Rooij, 2006) (Klinedinst, 2007), and (Franke, 2011)
[28] After all, *some* constructions of the form $\phi(A \vee B)$ convey $\phi(A)$ and $\phi(B)$ due to the semantics of ϕ: let ϕ be negation.

This effect is known as a *scalar implicature*. The hallmark of scalar implicatures is that the apparent entailment disappears (and even reverses) in negative (downward-entailing) environments. If (19a) entails (19b), the negation of (19b) should entail the negation of (19a). But this inference looks terrible:

(20) a. It's certain that it will rain.
b. $\not\rightsquigarrow$ It's not possible that it will rain.

The SDA effect passes this test:[29]

(22) a. I don't think the match would have been cancelled if it had rained.
b. $\not\rightsquigarrow$ I don't think the match would have been cancelled if it had rained or snowed.

Suppose $(A \vee B) > C$ entails $A > C$, in line with SDA. Then $\neg(A > C)$ should entail $\neg((A \vee B) > C)$, and thinking that $\neg(A > C)$ should incur a commitment to $\neg((A \vee B) > C)$. The inference from (22a) to (22b) should sound good. But it does not. (22b) conveys a belief that the match would have gone ahead in case of rain *and* in case of snow. That is, while $(A \vee B) > C$ appears to entail $A > C$ and $B > C$, $\neg((A \vee B) > C)$ appears to entail $\neg(A > C)$ and $\neg(B > C)$. No amount of tweaking the similarity analysis could explain this.

I have exploited the reversal of SDA under negation in the triviality arguments from section 4. It is not a coincidence that all my triviality arguments involved the contraposition of SDA – the inference from $\neg(A > C)$ to $\neg((A \vee B) > C$. While SDA has intuitive force, its contraposition does not.

All the constructions from the previous section ('might' counterfactuals, deontic conditionals, etc.) display an SDA-type effect in positive environments, and a reversed effect in negative environments.

Looking at the wider range of constructions leads to another argument for an implicature account. I said that there are many constructions ϕ for which $\phi(A \vee B)$ tends to convey both $\phi(A)$ and $\phi(B)$, even though this is not predicted by the standard semantics of these constructions. (We can now add $\neg((A \vee B) > C)$ to the list.) But there are also many constructions for which $\phi(A \vee B)$ does *not* tend to convey $\phi(A)$ and $\phi(B)$. 'I believe that A or B', for example, does not suggest that I believe both A and B. What makes the difference?

[29] Because plain negations of counterfactuals are generally unnatural, I here exploit the "neg-raising" effect of 'think', by which 'I don't think' comes to be interpreted as 'I think not'. If you prefer, you may substitute 'doubt' for 'I don't think'. The reversal of SDA can also be witnessed with quantified constructions:

(21) a. No animal would go to sleep if threatened or attacked.
b. $\rightsquigarrow$ No animal would go to sleep if threatened.
c. $\rightsquigarrow$ No animal would go to sleep if attacked.

The validity of SDA would predict an opposite inference from, for example, (21b) alone to (21a): by SDA, $\neg(Tx > Sx)$ entails $\neg((Tx \vee Ax) > Sx)$. So [Every $x : Ax$]$\neg(Tx > Sx)$ should entail [Every $x : Ax$]$\neg((Tx \vee Ax) > Sx)$.

Danny Fox has pointed out that the effect – whatever its explanation – appears to depend on the unavailability of a conjunctive alternative: $\phi(A \vee B)$ tends to convey $\phi(A)$ and $\phi(B)$ iff there is no alternative to $\phi(A \vee B)$ that expresses the conjunction of $\phi(A)$ and $\phi(B)$.[30]

To illustrate, let $\phi(A \vee B)$ be 'You may have tea or coffee'. There are, of course, sentences that express the conjunction of $\phi(A)$ and $\phi(B)$; for example, 'You may have tea and you may have coffee'. But this is (normally) not an alternative to 'You may have tea or coffee', in the semi-technical sense of 'alternative' (henceforth *scalar alternative*) that figures in theories of focus and scalar implicatures.[31] Loosely speaking, 'You may have tea and you may have coffee' is considerably more complex than 'You may have tea or coffee', and this disqualifies it as a scalar alternative. 'I believe that A or B', by contrast, has a conjunctive scalar alternative: 'I believe that A and B'. This is no more complex than 'I believe that A or B', and it is plausibly equivalent to 'I believe that A and I believe that B'. Accordingly, we get an SDA-type effect for 'May$(A \vee B)$', but not for 'Believe$(A \vee B)$'. We get the effect for counterfactuals because here, too, there is no relevant conjunctive alternative to $(A \vee B) > C$.

If Fox's conjecture is on the right track, the mechanism that gives rise to the conjunctive reading of $\phi(A \vee B)$ depends on the availability or unavailability of a conjunctive scalar alternative. This is just what we would expect if the mechanism is that of scalar implicatures.

For the case of counterfactuals, an implicature account also promises to shed light on the recalcitrant data from section 3. Recall, for example, the difference between (23a) and (23b).

(23) a. If Spain had joined the war, Hitler would have been pleased.
b. If Spain had joined the war on the side of the Allies or the Axis, Hitler would have been pleased.

(23b) suggests that Hitler would have been pleased even if Spain had joined the war on the Allied side; (23a) does not. The two antecedents are practically equivalent, but they have different alternatives: 'Spain joined the war on the side of the Allies' is an obvious alternative to the antecedent of (23b), but not to that of (23a).

All in all, there is strong evidence that the SDA effect is a scalar implicature. But how, exactly, does it arise? There is no shortage of proposals. (See, among others, (Loewer, 1976), (Nute, 1980a), (Bennett, 2003), (Klinedinst, 2007), (van Rooij, 2010), (Franke, 2011), (Bar-Lev and Fox, 2020), (Schwarz, 2021)[32].) I'll throw another proposal into the mix.

[30] See (Fox, 2007), (Singh et al., 2016), (Bar-Lev and Fox, 2020), (Fox and Katzir, 2021).
[31] See, for example, (Fox and Katzir, 2011) and (Breheny et al., 2018).
[32] I still have some hope for the proposal sketched at the end of (Schwarz, 2021). But I don't know how to make it work for 'might' conditionals or for 'would' conditionals with a similarity semantics. The proposal also doesn't seem to explain why the SDA effects disappear in negative environments.

6 A derivation

I'm going to assume, for the sake of the argument, that the similarity analysis is correct. We want to explain why an utterance of $(A \vee B) > C$ tends to convey that $A > C$ and $B > C$ are both true, even if one of A and B is more "remote" than the other.

The standard neo-Gricean recipe for deriving scalar implicatures compares the uttered sentence A with its scalar alternatives $\mathrm{Alt}(A)$. If a cooperative and well-informed speaker uttered A rather than some stronger alternative, we are supposed to infer that the alternatives must be false. Unfortunately, the most obvious alternatives to $(A \vee B) > C$ are $A > C$ and $B > C$, which are not stronger than $(A \vee B) > C$. Besides, we want to infer that they are true, not false. We have a little more work to do.

The key insight that helps us move further goes back to (Kratzer and Shimoyama, 2002). Kratzer and Shimoyama point out that a well-informed and cooperative speaker might prefer one utterance over another not only if the alternative is false, but also if it would merely have a false implicature.

Suppose our conversation turns to whether $A > C$ and whether $B > C$. In this context, an utterance of $A > C$ tends to implicate $\neg(B > C)$:

(24) X: Would the party have been fun if Alice or Bob had come?
Y: It would have been fun if Alice had come.
$\rightsquigarrow$ Not: It would have been fun if Bob had come.

This is probably an instance of a more general mechanism whereby propositions that are not affirmed are assumed to be denied:[33]

(25) X: Did Alice or Bob leave early?
Y: Alice left early.
$\rightsquigarrow$ Not: Bob left early.

The exact mechanism behind this "exhaustification" effect is, to my knowledge, an open question.[34] Let's simply take for granted that in a context where a speaker has uttered $(A \vee B) > C$, each of $A > C$ and $B > C$ would typically implicate the negation of the other.

Now imagine you know that $A > C$ and $B > C$ are both true, and you want to communicate this to your addressee. More precisely, suppose you want to increase your addressee's degree of belief in $(A > C) \wedge (B > C)$. You could utter $(A > C) \wedge (B > C)$. But suppose you also have a penchant for the pithy. You hate laborious expressions. You could utter just $A > C$, or just $B > C$. These are nice and short, but they would not lead to accurate beliefs, as either implicates the negation of the other. (Or so we assume.) The optimal choice

[33] The implicature from $A > C$ to $\neg(B > C)$ is also closely related to the phenomenon of "Conditional Perfection" ((Geis and Zwicky, 1971)), whereby $A > C$ tends to implicate $\neg(\neg A > C)$.

[34] See, for example, (Schulz and Van Rooij, 2006) and (Cremers et al., 2023).

might be $(A \vee B) > C$. This is still reasonably short, and it should increase the addressee's confidence in $(A > C) \wedge (B > C)$, if they believe what you say.[35]

Now let's switch perspectives. Imagine you're the addressee and you receive the message $(A \vee B) > C$. You believe the speaker to be rational, cooperative, and well-informed about $A > C$ and $B > C$. What information might they have? They can't know that $A > C$ and $B > C$ are both false, as this would contradict the literal content of their utterance. Could they know that $A > C$ is true and $B > C$ false? It seems not. In that case, $A > C$ would have been shorter and more informative (due to the exhaustification effect). By the same reasoning, the speaker can't know that $B > C$ is true and $A > C$ false. The only remaining possibility is that $A > C$ and $B > C$ are both true. As we've just seen, a speaker who knows this may indeed utter $(A \vee B) > C$, if they have a strong enough preference for brevity.[36]

At this stage, we can already predict the SDA inference. But we also predict that an utterance of $(A \vee B) > C$ conveys a strong preference for brevity. We can get rid of this unwanted second prediction by switching perspectives once more.

Imagine again that you are the speaker and you'd like to convey your knowledge of $A > C$ and $B > C$. You wonder what your addressee will infer from an utterance of $(A \vee B) > C$. If they will go through the line of reasoning just described, they will infer that $A > C$ and $B > C$ are both true. This is just what you want to achieve. It follows that you may utter $(A \vee B) > C$ even if you do not have a strong preference for brevity.

Finally, if you are an addressee who believes that the speaker has gone through *this* line of reasoning, you will take an utterance of $(A \vee B) > C$ to signal knowledge of $A > C$ and $B > C$, and you will no longer infer that they have a strong preference for brevity.

What's up with the iterated switch of perspectives? It can't be found in (Grice, 1975). But it has a motivation. Grice assumes that speakers tend to be rational and cooperative, and that hearers can draw on this assumption to interpret an utterance. He didn't realize that this leads to a regress.[37] In order to choose an utterance, a speaker needs to anticipate how the hearer would respond. The hearer's response, however, turns on their reconstruction of the speaker's decision problem. The speaker needs a model of the hearer that includes a model of the speaker, and so on.

Since our cognitive resources are limited, we can expect these models within models to be increasingly unsophisticated, bottoming out in a "literal" hearer

[35] This follows from the fact that $(A > C) \wedge (B > C)$ entails $(A \vee B) > C$ (according to the similarity analysis and almost all of its rivals). By probability theory, $P(H/E) > P(H)$ whenever H entails E, provided that $P(H) > 0$ and $0 < P(E) < 1$.

[36] Suppose the hearer doesn't think the speaker has a preference for simplicity. If the speaker knows that $A > C$ and $B > C$ are both true and wants to convey this information, they would then utter $(A > C) \wedge (B > C)$, rather than the strictly less informative but simpler $(A \vee B) > C$. The utterance of $(A \vee B) > C$ therefore signals that either the speaker has a preference for simplicity or doesn't want to convey the full truth about whether $A > C$ and $B > C$ (perhaps because they don't know the full truth).

[37] (Lewis, 1969) saw it. See, esp., pp.27ff.

or a "literal" speaker who does not engage in strategic reasoning.

You may doubt that we engage in *any* kind of strategic reasoning when we interpret simple utterances like (1a). This may be right. My suggestion is only that we interpret utterances *as if* we were engaged in such reasoning. In practice, we may well have learned to draw the relevant inferences automatically, at least if nothing alerts us to the falsity of their premises.

The model of communication I have outlined is formalized in the "Rational Speech Act" (RSA) framework. (See (Scontras et al., 2021).) With a bit of tidying, the above derivation can be spelled out as an RSA model. I won't descend into the formal details here.[38] But I want to flag one point that needs tidying.

I claimed that a well-informed speaker with a sufficiently strong preference for brevity would choose $(A \vee B) > C$ only if they know that $A > C$ and $B > C$ are both true. A speaker who knows that $A > C$ is true and $B > C$ false, for example, would prefer $A > C$ over $(A \vee B) > C$, as it conveys more information *about* $A > C$ *and* $B > C$. By the similarity analysis, however, $(A \vee B) > C$ carries information not just about $A > C$ and $B > C$, but also about the relative closeness of A and B. The speaker might prefer $(A \vee B) > C$ over $A > C$ because they want to convey this extra information.

But this seems to be unusual. In a normal situation, when a speaker utters $(A \vee B) > C$, they intend to convey information on whether $A > C$ and whether $B > C$; the relative closeness of A and B is not of interest. We can use this datum to fix the derivation. The norms of conversation only require speakers to be informative with respect to questions that are of interest ("under discussion"). In normal contexts, the fact that $(A \vee B) > C$ carries information about the relative closeness of A and B is therefore no reason to prefer it.

The question-sensitivity of the derivation might shed light on the puzzle that (26b) becomes more acceptable if it follows (26a), as noted by Nute:

(26) a. If Spain had fought with the Allies or the Axis, it would [probably] have fought with the Axis.
b. If Spain had fought with the Allies or the Axis, Hitler would [probably] have been pleased.

(26a) is most naturally interpreted as conveying information about the relative closeness of the *Allies* and the *Axis* possibilities.[39] Once this question has been raised, it might still be salient when we interpret (26b).

Looking back at the other data from the previous sections, the derivation I have outlined explains why SDA effects are sensitive to substitution of equivalents and why they disappear under negation. Since it is not based on doctoring with the meanings of '>' and 'or', the derivation smoothly generalizes to other

[38] My derivation loosely resembles the derivation of Free Choice in (Champollion et al., 2019).

[39] My derivation of SDA does not go through for (26a) because $\mathit{Axis} > \mathit{Allied}$ can hardly be exhaustified to imply the negation of $\mathit{Allied} > \mathit{Allied}$. More generally, it is implausible that a speaker of (26a) wants to convey information about whether $\mathit{Axis} > \mathit{Allied}$ and whether $\mathit{Allied} > \mathit{Allied}$.

constructions. For example, to explain why $(A \vee B) >_m C$ tends to convey $(A >_m C) \wedge (B >_m C)$, we would assume that $A >_m C$ is normally exhaustified to implicate $\neg(B >_m C)$. A cooperative and well-informed, but not overly sophisticated speaker would therefore choose $(A \vee B) >_m C$ only if they know that $A >_m C$ and $B >_m C$ are both true and they have a strong preference for brevity. As before, a more sophisticated speaker might use $(A \vee B) >_m C$ even without a strong preference for brevity. (The triviality arguments from section 4 fail because the derivations do not go through under negation.)

What about unspecific antecedents that aren't (binary) disjunctions? Suppose an antecedent A divides into two sub-possibilities, A_1 and A_2.[40] Unlike the disjuncts of a disjunction, these do not automatically qualify as scalar alternatives to A and to one another. But they plausibly do so if they are sufficiently salient in the conversational context.[41] In that case, one can run essentially the same derivation as above to show that $A > C$ implicates $A_1 > C$ and $A_2 > C$.

What if the antecedent divides into more than two salient sub-possibilities $A_1, A_2, \ldots, A_n$? If all conjunctions (unions) of these possibilities are also salient, we can predict that $A > C$ implicates $A_i > C$ for all A_i.[42] If no intermediate possibilities are salient, the account may only predict an inference that $A_i > C$ is true for *multiple* A_i.[43] In (Schwarz, 2021), I argued that this "partial" or "existential" prediction is correct for Free Choice. 'Fred might be in France', for example, tends to convey that there are multiple locations in France where Fred might be, but not that every location in France is a possibility. Whether this prediction is also true for counterfactuals is not clear to me.[44]

As I said, I don't want to go too far into the details of the proposed derivation. I have introduced it only as a proof of concept, as an illustration of how SDA effects might be explained as a kind of implicature. All I want to suggest is that this direction looks more promising than trying to derive the effects by

[40] That is, assume that A is (contextually) equivalent to $A_1 \vee A_2$, and that $A_1 \wedge A_2$ is (contextually) ruled out. When I talk of "possibilities", I equivocate, for the sake of simplicity, between expressions and their meanings.

[41] This is a common assumption in the literature; see, for example, (Katzir, 2007) and (Breheny et al., 2018).

[42] We need to assume that if A' is an alternative that excludes some A_j, then $A' > C$ would be exhaustified to imply $\neg(A_j > C)$. A speaker who knows $\bigwedge_i A_i > C$ would therefore prefer $A > C$ over $A' > C$. A speaker who knows that $A' > C$ and $\neg(A_j > C)$, by contrast, would prefer $A' > C$. So $A > C$ conveys that $A_i > C$ for all A_i.

[43] In this case, $A > C$ might also be the optimal choice for a speaker who knows, say, $A_1 > C \wedge \ldots \wedge A_{n-1} > C \wedge \neg A_n > C$. It would not, however, be optimal for a speaker who knows that a single $A_i > C$ is true and the rest are false, since $A_i > C$ would then convey more relevant information.

[44] If the sub-possibilities $A_1, \ldots, A_n$ are equally close, similarity semantics alone implies that $A > C$ can only be true if each $A_i > C$ is true. A test case would have to involve sub-possibilities with different degrees of remoteness. We also have to ensure that no intermediate possibilities are salient. Both of these are somewhat hard to assess independently.

tweaking the semantics of counterfactuals.[45][46]

To be clear, this needn't be bad news for Hájek's chance account. It *does* undercut one motivation for the account – the account's ability to explain SDA effects. But I've argued that this was anyway an illusion: the chance account does not provide a satisfactory explanation of SDA data. No version of the similarity analysis does. But that's OK. If the data can be explained as an implicature, the chance account is off the hook.

7 The case for skepticism

Let's return to the case for counterfactual skepticism. The argument from unspecific antecedents, you may recall, starts with the observation that ordinary counterfactuals often have unspecific antecedents. It assumes that a counterfactual with an unspecific antecedent is true only if it remains true on every way of resolving the unspecificity. Since one can often find resolutions that make the counterfactual false, the argument concludes that most ordinary counterfactuals are false.

I have cast doubt on the central premise: that a counterfactual with an unspecific antecedent is true only if it remains true on every way of resolving the unspecificity. This assumes that a generalized form of SDA is semantically valid. I have argued that it is not. Granted, counterfactuals with unspecific antecedents often *appear* to entail corresponding counterfactuals with more specific antecedents. But the appearance is misleading. The effect is better explained as an implicature.[47]

Let's have another look at the example from section 1. Bob is on a medication that would cause an allergic reaction if he consumed alcohol. In this context, (27) seems true.

(27) If Bob had wine, his medication would cause an allergic reaction.

The skeptic now raises the possibility of de-alcoholized wine: if Bob had de-alcoholized wine, his medication would not cause a reaction. Once this possibility is raised, (27) looks problematic.

The implicature account explains why (27) might look problematic, even if it is true. 'Bob has wine' divides into 'Bob has regular wine' and 'Bob has de-alcoholized wine'. If both of these are contextually salient, (27) tends to implicate that in either case, Bob's medication would cause an allergic reaction.

[45] (Fox and Katzir, 2021) complain that RSA-type accounts of Free Choice and SDA are implausibly sensitive to the agents' priors, and that they fail to account for the effect with more than two disjuncts. This is true for the account of (Franke, 2011) and the toy model constructed by Fox and Katzir, but not of the model I have sketched (nor of that in (Champollion et al., 2019)).

[46] Yet other proposals could, of course, be explored. An important third direction expands on the idea that 'or' may not simply express Boolean disjunction. See, for example, (Alonso-Ovalle, 2008), (Ciardelli et al., 2018), (Romoli et al., 2022).

[47] The same kind of response is available to (Loewenstein, 2021)'s argument from reverse Sobel sequences.

But this is not a semantic entailment. If the de-alcoholized wine possibilities are more "remote", (27) is true.

My response to the skeptical argument is contextualist, but different from standard contextualist moves. A more standard contextualist might say that de-alcoholized wine possibilities become close by being made salient Alternatively, they might say that a counterfactual is true as long as all the *relevant* closest A worlds are C worlds, and that de-alcoholized wine worlds are normally irrelevant.[48] I'm suggesting that it does not matter if de-alcoholized wine worlds are remote or close. (I suspect this depends on the details of the scenario.) Either way, attending to these worlds will tend to associate (27) with a false implicature.

There are, of course, other arguments for counterfactual skepticism. There is the argument from chance (see (Hájek, 2021a) or (Hájek, 2023)). Take (28):

(28) If I had put an ice cube in my tea, it would have melted.

Quantum physics tells us that a state in which an ice cube has been put into my tea has a positive chance of evolving into a state in which the entire liquid is frozen. This suggests that the ice cube *might not* have melted. And doesn't this, in turn, suggest that (28) is false?

The argument from chance does not seem to rely on the unspecificity of the antecedent, so the implicature account may have little to say about it.[49] The account does, however, have something to say about a version of the argument that appeals to *statistical* mechanics rather than quantum mechanics. Classical statistical mechanics also assigns a positive probability to the tea freezing, but not because it posits an indeterministic dynamics. According to statistical mechanics, the melting of an ice cube in a hot liquid depends on the exact configuration of all relevant molecules. Some configurations would not lead to a melting. If these configurations are no more remote than more ordinary ones, (28) is predicted to be false.

Conversely, if we want to hold on to (28), we have to say that the freezing configurations are more remote. Remember that remoteness is a technical concept, not to be judged by intuitions about distance or similarity. A skeptic might insist that the freezing configurations can't be more remote because attending to them makes (28) seem false. The implicature account defuses this objection. It predicts that attending to the peculiar configurations can make (28) seem false, even if it is true.

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